

Kyle Norkett

(226) 332-5517 · Email: knorkett@protonmail.com · Github: [@CryptoFowl](https://github.com/CryptoFowl)

Smart Contract / Protocol Engineer specializing in secure on-chain systems, mechanism design, and production Solidity. 4+ years building and deploying contracts across Ethereum and L2s with experience spanning protocol architecture, economic modeling, and security-focused development. Comfortable owning systems end-to-end from design through deployment & audit.

CORE SKILLS

Smart Contract and EVM:

Solidity • Foundry • Hardhat • EVM internals • Contract architecture

Protocol & Security:

Protocol design • Invariants & fuzzing • Governance • Economic modeling • Threat modeling

Supporting:

TypeScript • Python • Noir (ZK) • On-chain analytics

PROFESSIONAL EXPERIENCE

Civitai - *Smart contract developer (2025)*

- Designed and deployed on-chain payment and billing infrastructure for an AI image generation platform
- Advised on operational security practices and contract risk mitigation
- Implemented and tested integration with ZK-based on-ramp tooling

Theopetra Labs - *Lead smart contract developer (2021-2024)*

- Led development of core protocol contracts across Ethereum, zkSync, and Base, owning architecture, testing strategy, and deployments
- Identified and resolved two critical vulnerabilities during pre-deployment review, preventing potential loss of funds
- Designed escrow system for automated real estate transactions using optimistic oracles and ZK proofs
- Reduced payment settlement costs $\sim 10\times$ through L2 migration strategy
- Migrated ERC-20 token and liquidity infrastructure to Base to support user growth
- Built financial simulations and pricing models to evaluate protocol economics and risk
- Developed tooling and dashboards to analyze on-chain activity and user behavior
- Partnered with product and operations to define protocol policies and security practices

PROJECT EXPERIENCE

D-Risk: Liquidation Cascade Simulation Engine (2025–Present)

- Designed and built a DeFi liquidation cascade simulation engine for stress-testing Ethereum lending markets and modeling forced-flow dynamics during market crashes
- Modeled recursive liquidation reflexivity across Aave-style collateral systems, including liquidation sequencing, correlated collateral exposure, eMode handling, oracle TWAP lag, liquidity withdrawal, and cross-market contagion dynamics
- Integrated live router-based price impact estimation using forked Uniswap liquidity to simulate executable slippage and liquidity exhaustion under stress conditions
- Developed historical crash replay and backtesting framework covering major crypto deleveraging events with configurable volatility regimes and liquidation-flow sensitivity analysis
- Built tooling to reconstruct on-chain lending positions and evaluate cascade amplification effects under varying market conditions

ProofLayer (2026) - Smart Contract & ZK Developer

- Built privacy-preserving KYC system using zero-knowledge proofs
- Implemented Noir circuit proving Merkle inclusion without revealing user data
- Enabled reusable credentials for attribute verification without exposing PII

EIP-7702 Subscription Payments (2025) - Smart contract developer

- Designed subscription primitive eliminating unlimited token approvals via EIP-7702
- Implemented lifecycle controls enabling recurring payments with only user signatures after initial setup

OPEN SOURCE CONTRIBUTIONS

DeFi Llama (2025-2026)

Open source data aggregator for decentralized finance

- Added liquidation data for AAVE v3 instances across multiple chains
- Updated schema and liquidation thresholds to include efficiency-mode data
- Added MetaLeX TVL adapter to track escrowed balances for CyberCORPs

Project Lighthouse (2022)

Clean-room, open-source custom server for LittleBigPlanet

- Applied cryptanalysis to improve security, allowing for authentication using PSN servers
- Improved moderation features

EDUCATION

University of Guelph

Guelph, ON

Bachelor of Arts in Sociology, Minor in Economics
